

# Conditional diffusion emulator of Veros ACC equilibrium states: first end-to-end results

21 September 2026

## Summary

- Goal: generate the post-spin-up  $(T, S)$  state of the Veros ACC channel directly from the EKE-closure coefficients  $(c_k, c_\epsilon)$ , skipping spin-up.
- Pipeline: DINO-Fusion diffusion code adapted to a parameter ensemble; runs end to end on Jean Zay in under one A100 hour.
- Result: with per-level normalisation the model's hold-out error is  $\approx$  the mean-state baseline (0.15 K); it beats the nearest training run only where the hold-out gap is large, and ties copying the row below when extrapolating one row. A Wasserstein metric on the stratification profiles gives the same ranking and shows that training and held-out conditions score alike.
- Per-level min-max normalisation (data range  $\rightarrow [-1, 1]$ ), tested on the same three splits, is worse (0.20 / 0.23 / 0.29 K): the ensemble spread halves and the bias dominates (Table 1).
- Physical miss: the model generates temperature inversions at the rate of the weakly eddying runs in every regime.

## 1 Data

- 100 Veros ACC runs,  $10 \times 10$  log-spaced grid:  $c_k \in [0.0125, 0.8]$ ,  $c_\epsilon \in [0.0875, 5.6]$ ; 100 years, 30-day output, interior grid  $30 \times 42 \times 15$  (x, y, z).
- Kept:  $T$  and  $S$ , last 20 years (241 snapshots/run).  $S = 35$  psu everywhere in every run.
- Nearly deterministic end states: within-window std 0.01 K above 1400 m vs 0.05 K between grid neighbours; bottom level still drifts by 0.45 K over the window.
- One effective parameter:  $\log c_k - 1.05 \log c_\epsilon$  explains 99 % of the variance of the domain-mean  $T$ ; the low- $c_k/c_\epsilon$  part of the grid saturates.
- Per-level distribution of  $T$  (Figure 1): above 400 m a mixture of restoring values and interior, with  $\sigma_z$  set by the meridional gradient, so  $\mu_z \pm 3\sigma_z$  is wider than the level and the normalised values stay in  $[-0.7, 0.4]$ ; below 1400 m a warm tail to  $x' = +2.7$ . Run-to-run signal  $\sigma_z^{\text{runs}}$ : 0.7 % of  $3\sigma_z$  at the surface, 26 % at the bottom.

## 2 Method

- Fields  $\rightarrow$  30 channels  $(T_{1..15}, S_{1..15})$ , normalised per level, land  $\rightarrow 0$ , zero-padded  $42 \times 30 \rightarrow 48 \times 32$ :

$$x'_z = \frac{x_z - \mu_z}{3 \max(\sigma_z, 0.05)}. \quad (1)$$

$\mu_z, \sigma_z$  are computed once on all 100 runs and held fixed for every hold-out split (a deliberate, mild leakage of 30 scaling constants; no held-out state enters training). The sampler clips the predicted clean state at  $|x'| \leq 1$  (three standard deviations) at every step. Alternative tested: per-level min-max,  $x'_z = 2(x_z - \min_z)/(\max_z - \min_z) - 1$ , with land and padding cells at the normalised level mean instead of 0 (Table 1, last rows).

Distribution of  $T$  per level, all 100 runs  $\times$  241 snapshots, water cells (surface at the top). Black:  $\mu_z$ ; red dashed:  $\mu_z \pm 3\sigma_z$ ; orange ticks: the 100 per-run level means.

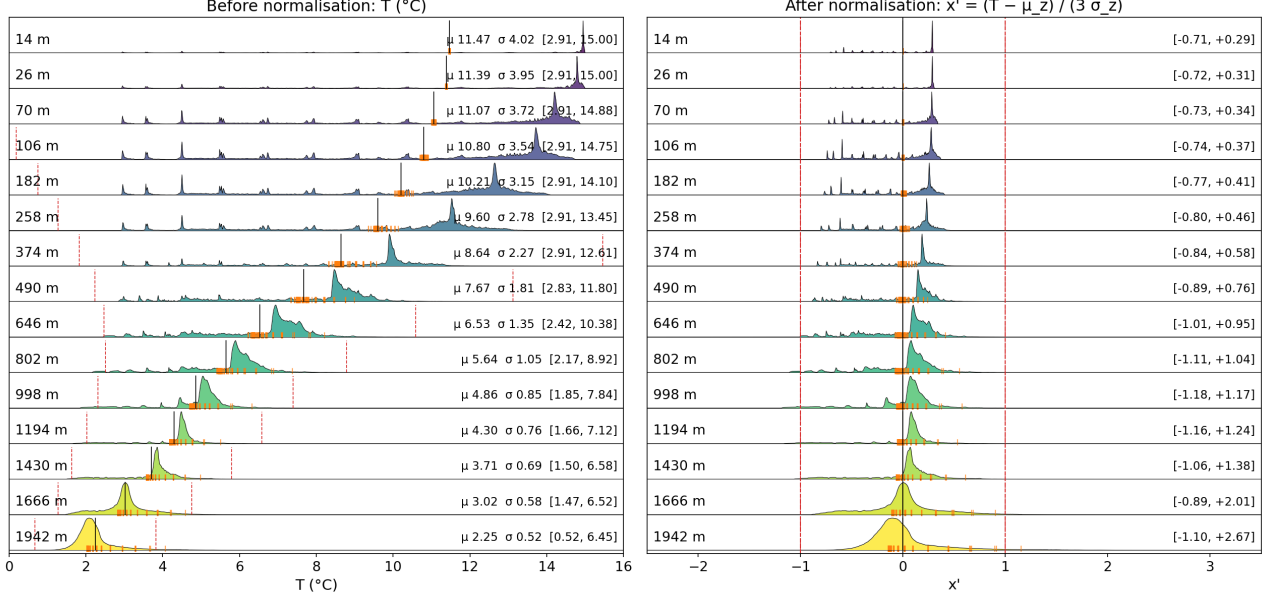


Figure 1: Distribution of  $T$  per level over all 100 runs and 241 snapshots (water cells), before (left) and after (right) the per-level normalisation of Eq. (1). Black:  $\mu_z$ ; red dashed:  $\mu_z \pm 3\sigma_z$ ; orange ticks: the 100 per-run level means.

- Denoiser: UNet2DModel (64, 64, 128, 128), 6.9 M parameters. The condition  $c$ , the standardised pair  $(\log c_k, \log c_\epsilon)$ , enters through an MLP added to the timestep embedding.
- DDPM,  $\epsilon$ -prediction, 1000 steps, squared-cosine schedule, EMA:

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, \epsilon} \left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t, c) \right\|^2. \quad (2)$$

20 000 steps, batch 32, AdamW  $10^{-4}$  with cosine decay; 22 min on one A100.

- Sampling: 1000-step DDPM loop with EMA weights; land and padding re-zeroed after every step; 32 samples per condition.

### 3 Evaluation

- Truth: 20-year time-mean state of each held-out run; metrics on water cells.
- Baselines: mean training state; nearest training run in standardised log-parameter space; average of the axis-neighbour training runs on the grid.
- Three hold-out splits, same data file and statistics, different training sets: scattered (10 interior points), band of three rows ( $c_k = 0.126, 0.2, 0.3175$ ; 30 runs; training rows a factor 6.3 apart), top row ( $c_k = 0.8$ ; 10 runs; extrapolation).
- Distributional metric. Each state is reduced to its profile of horizontal means over water cells,  $\bar{T}_z$ . Per level, the Wasserstein-1 distance between the 32 generated values and all 241 true snapshots of the window, through the quantile functions,

$$W_1^{(z)} = \int_0^1 |F_{g,z}^{-1}(q) - F_{t,z}^{-1}(q)| dq, \quad W_1 = \sum_z \frac{\Delta z_z}{H} W_1^{(z)}, \quad (3)$$

with  $\Delta z_z$  the level thicknesses from the level midpoints ( $H = 2080$  m). All 241 snapshots are used, not a subsample of 32: the quantile form accepts unequal sizes and a subsample would only add noise. A point prediction  $x$  (the baselines) gives  $W_1^{(z)} = \frac{1}{241} \sum_i |x_z - \bar{T}_{z,i}|$ . Unlike the RMSE of the ensemble mean, this rewards a correct spread and does not reward collapse to the mean; it ignores horizontal structure, which the RMSE keeps.

|                                                | scattered<br>10 runs | band of three rows<br>30 runs | top row<br>10 runs |
|------------------------------------------------|----------------------|-------------------------------|--------------------|
| diffusion, ensemble mean of 32: RMSE (K)       | 0.152                | 0.135                         | 0.173              |
| diffusion, single sample: RMSE (K)             | 0.188                | 0.178                         | 0.214              |
| nearest training run: RMSE (K)                 | 0.037                | 0.116                         | 0.170              |
| neighbour average: RMSE (K)                    | 0.020                | 0.116                         | 0.170              |
| mean training state: RMSE (K)                  | 0.155                | 0.197                         | 0.470              |
| diffusion: domain-mean bias (K)                | −0.027               | −0.027                        | −0.071             |
| ensemble spread / true spread (K)              | 0.109 / 0.031        | 0.111 / 0.031                 | 0.115 / 0.032      |
| $T$ inversions (interfaces), generated / truth | 9.6 % / 7.7 %        | 10.7 % / 1.3 %                | 12.1 % / 0.1 %     |
| max salinity error (psu)                       | 0.011                | 0.013                         | 0.018              |
| $W_1$ profile, diffusion (K)                   | 0.127                | 0.111                         | 0.143              |
| $W_1$ profile, neighbour average (K)           | 0.036                | 0.095                         | 0.142              |
| $W_1$ profile, nearest training run (K)        | 0.044                | 0.095                         | 0.142              |
| $W_1$ profile, mean training state (K)         | 0.136                | 0.160                         | 0.420              |
| min-max norm., ensemble mean of 32: RMSE (K)   | 0.199                | 0.233                         | 0.288              |
| min-max, single sample: RMSE (K)               | 0.205                | 0.238                         | 0.293              |
| min-max, ensemble spread (K)                   | 0.051                | 0.056                         | 0.061              |
| min-max, $W_1$ profile (K)                     | 0.165                | 0.188                         | 0.241              |

Table 1: Hold-out metrics per split, mean over the held-out runs, water cells only, 32 samples per run, 3-std normalisation. Last four rows: the same splits trained with the per-level min-max normalisation.

## 4 Findings

- **Conditioning is weak under per-level scaling.** Signal-to-range ratio  $\rho_z = \sigma_z^{\text{runs}} / (3\sigma_z)$  is 0.7 % at the surface, 9 % at 650 m, 25 % at the bottom; the  $\epsilon$ -loss weights all cells equally, so the shared structure dominates. Scattered split: RMSE = mean-state baseline (Figure 2).
- **Model error is split-independent (0.14–0.17 K); baselines degrade with the gap (0.02–0.17 K).** The model beats the nearest run in the upper band row (0.127 vs 0.152 K), ties the middle one, loses the lower one, and beats copying the row below over the trending part of the top row ( $c_e$  0.14–0.56: 0.12–0.23 vs 0.18–0.31 K) while losing at the corner and in the flat regime (Figures 3, 4).
- **Depth structure.** Beats the nearest run between 370 and 800 m in the band split (0.08 K vs 0.14 K at 650 m) and between 260 and 1430 m in the top row; worse than baselines in the top 250 m (scale  $3\sigma_z \approx 12$  K amplifies sampler noise) and at the two bottom levels (skewed, drifting training distribution).
- **The sampler’s clip is a regulariser, not a range limit.** Same trained models, sampling only: clip at  $|x'| \leq 1$  gives 0.152 / 0.173 K (scattered / top row), at 1.5 it gives 0.194 / 0.183 K, at 3 it gives 0.302 / 0.277 K, although the final samples barely exceed  $|x'| = 1$  (0.1 % of values). Clipping the early, inaccurate clean-state estimates keeps the chain on track. The price is a range limit for extrapolation: with the fixed statistics 12 % of the held-out top row’s true bottom values lie above  $\mu_z + 3\sigma_z$  and cannot be generated.
- **Min-max normalisation is worse on every split** (0.199 / 0.233 / 0.288 K against 0.152 / 0.135 / 0.173;  $W_1$  0.165 / 0.188 / 0.241 against 0.127 / 0.111 / 0.143) and at almost every level: the 100–400 m band doubles (0.13  $\rightarrow$  0.24 K at 106 m, scattered) and the bottom level goes 0.34  $\rightarrow$  0.43 / 0.28  $\rightarrow$  0.54 / 0.35  $\rightarrow$  0.66 K. The ensemble spread halves (0.05 vs 0.11 K) and a single sample scores like the ensemble mean: the error is bias, not sampler noise. Stretching the upper levels onto  $[-1, 1]$  gives the fixed structure of Figure 1 more weight, not the run-to-run signal (1.4 % of the scale); stretching the bottom scale to the corner cell dilutes the level that carries the most signal.
- **Inversions.** Generated 10–12 % of interfaces with  $\partial_z T < 0$  in every regime; truth: 7.7 % (scattered)  $\rightarrow$  1.3 % (band)  $\rightarrow$  0.1 % (top). The regime dependence is not learned.
- **$W_1$  on the mean profiles agrees with the RMSE ranking and adds two facts** (right-hand panels of Figures 2–4). (i) Training grid points score the same as held-out ones (0.124 vs 0.127 K, scattered split): the model does not fit the training conditions either, the signature of the mean-

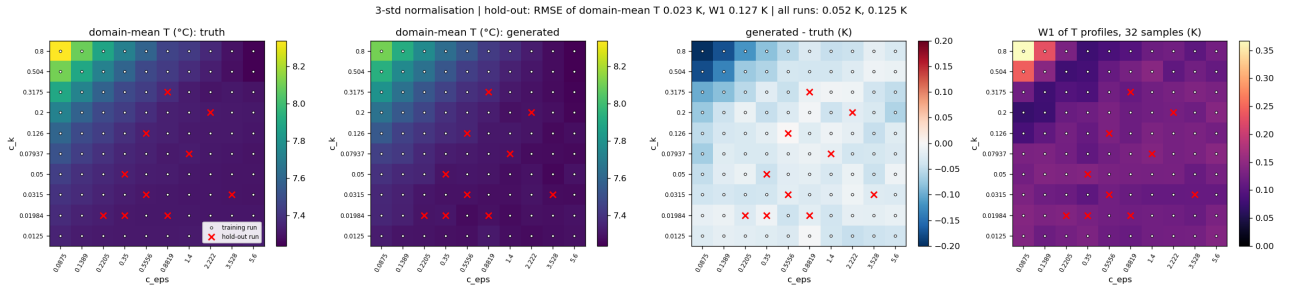
state collapse. (ii) Around 650 m the samples beat the baselines under  $W_1$  in the band and top splits (0.06 vs 0.12 and 0.08 vs 0.17 K), because a point prediction is charged for the spread it cannot represent; the model’s total is set by the top 250 m and the bottom two levels, as for the RMSE.

- **Plumbing.**  $S$  within 0.02 psu of 35 with no constraint; land exact; ensemble spread  $3.5\times$  the true within-window spread (Figures 5–7).

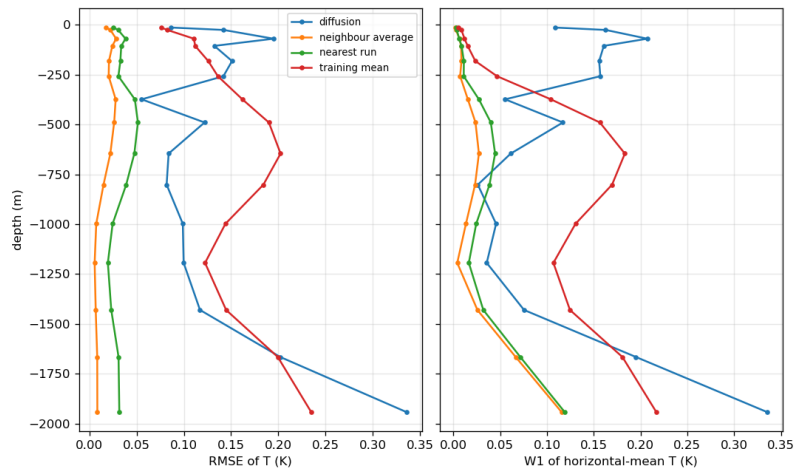
## 5 Next steps

- Stratification constraint at sampling time (DINO-Fusion’s isotonic projection, on  $T$  since  $S$  is constant): targets the inversion miss directly.
- Stronger conditioning: classifier-free guidance; year as a third condition (drift of 0.45 K at the bottom inside the window).
- Evaluate against snapshots as well as the time mean (nearest-snapshot RMSE, spread–skill).
- Then: same pipeline on a global Veros configuration (active  $S$ , seasonal cycle, multi-century spin-up).

**Reproducibility.** Branch `miniveros` of the DINO-Fusion repository, package `miniveros/`; commit `5fdcf01` for the three runs reported here (training at `2b542aa`, sampling with the clip range of `5fdcf01`; min-max runs at `4fe5d51`). Extraction 2.5 min (8 cores), training 17–22 min (one A100), generation of the hold-out and grid sets + evaluation 14–18 min.

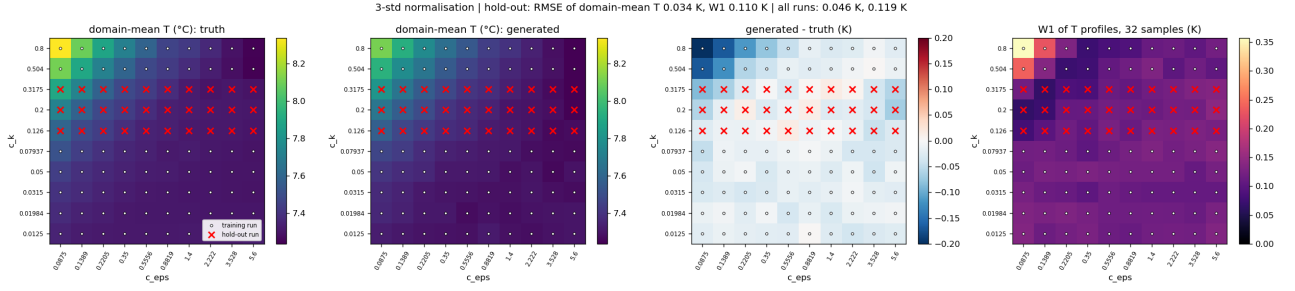


(a) Over the parameter grid: domain-mean  $T$  of the truth and of the generated states, their difference, and  $W_1$  of the profile distribution. White dots: training runs; red crosses: held-out runs.

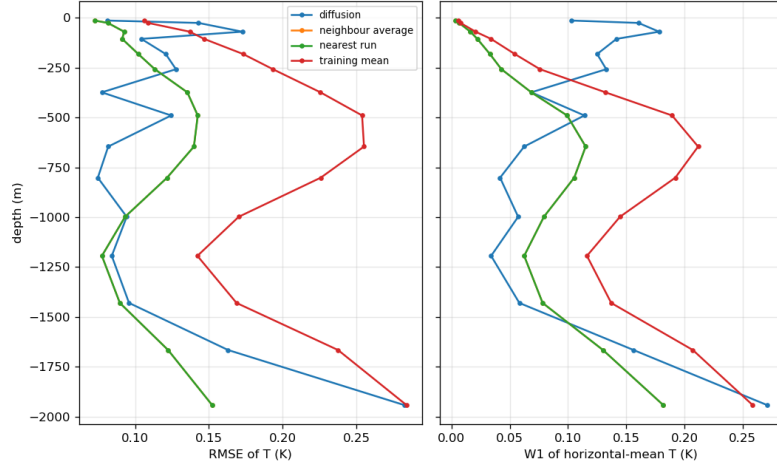


(b) RMSE (left) and  $W_1$  (right) by depth, mean over the held-out runs.

Figure 2: Scattered split (10 interior held-out runs), 32 samples per run.

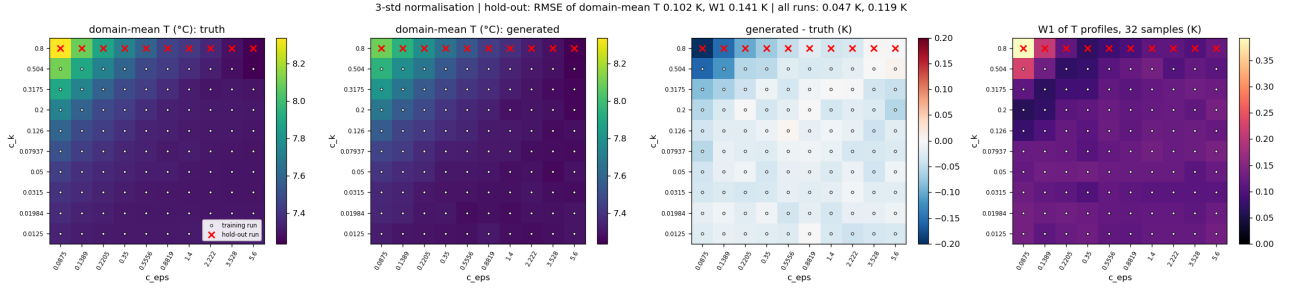


(a) Over the parameter grid: domain-mean  $T$  of the truth and of the generated states, their difference, and  $W_1$  of the profile distribution. White dots: training runs; red crosses: held-out runs.

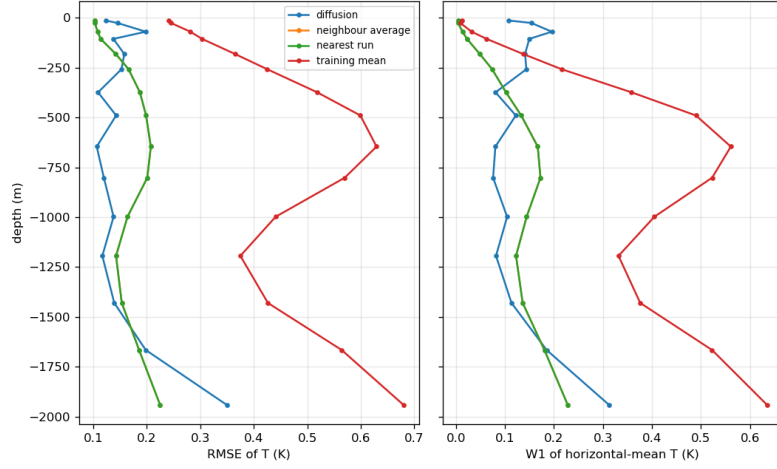


(b) RMSE (left) and  $W_1$  (right) by depth, mean over the held-out runs.

Figure 3: Band of three rows ( $c_k = 0.126, 0.2, 0.3175$  held out), 32 samples per run.



(a) Over the parameter grid: domain-mean  $T$  of the truth and of the generated states, their difference, and  $W_1$  of the profile distribution. White dots: training runs; red crosses: held-out runs.



(b) RMSE (left) and  $W_1$  (right) by depth, mean over the held-out runs.

Figure 4: Top row ( $c_k = 0.8$  held out, extrapolation), 32 samples per run.

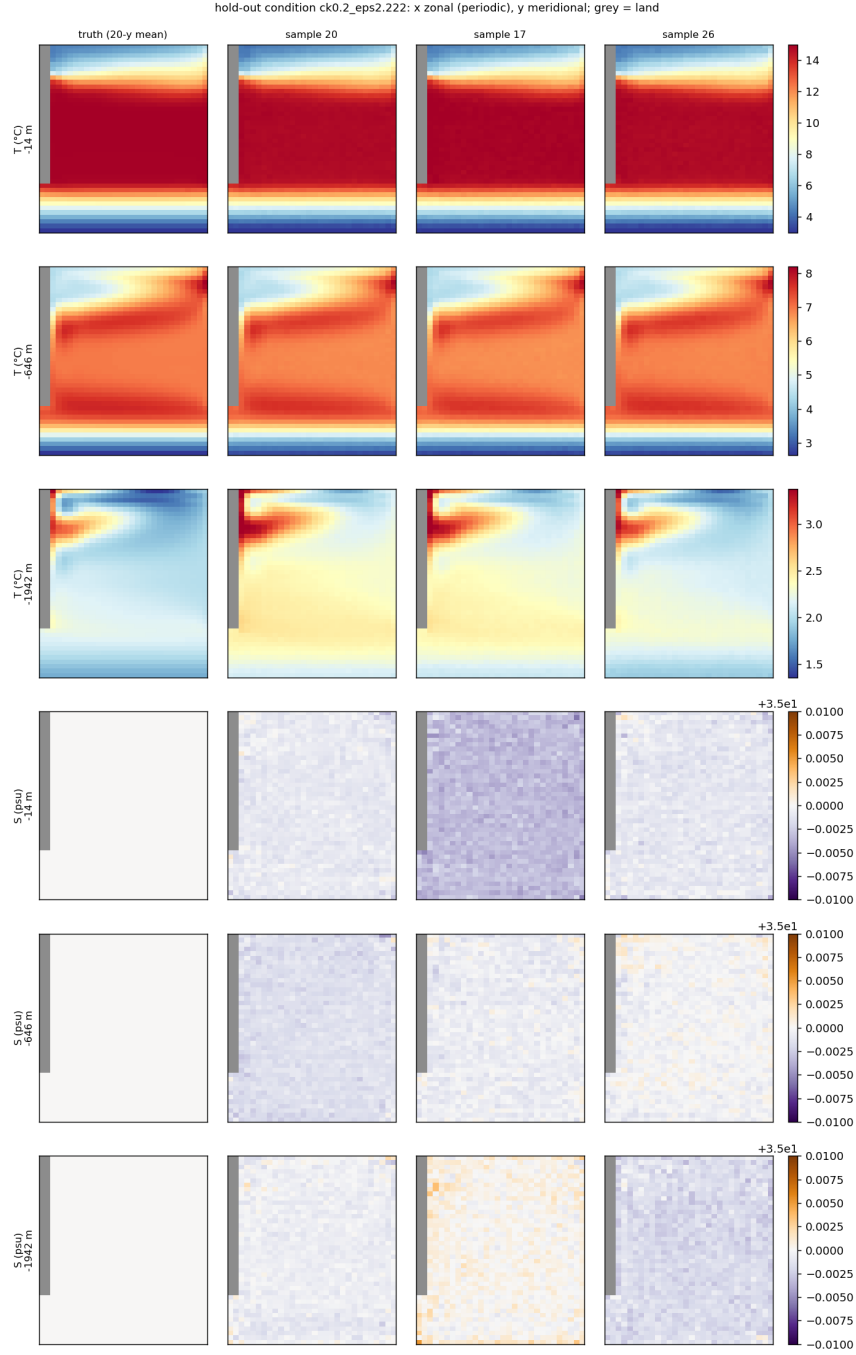


Figure 5: Scattered split,  $c_k = 0.2$ ,  $c_\epsilon = 2.222$ : true 20-year mean (left) and three random samples at the surface, 650 m and the bottom;  $T$  (top three rows),  $S$  (bottom three rows). Grey: land.

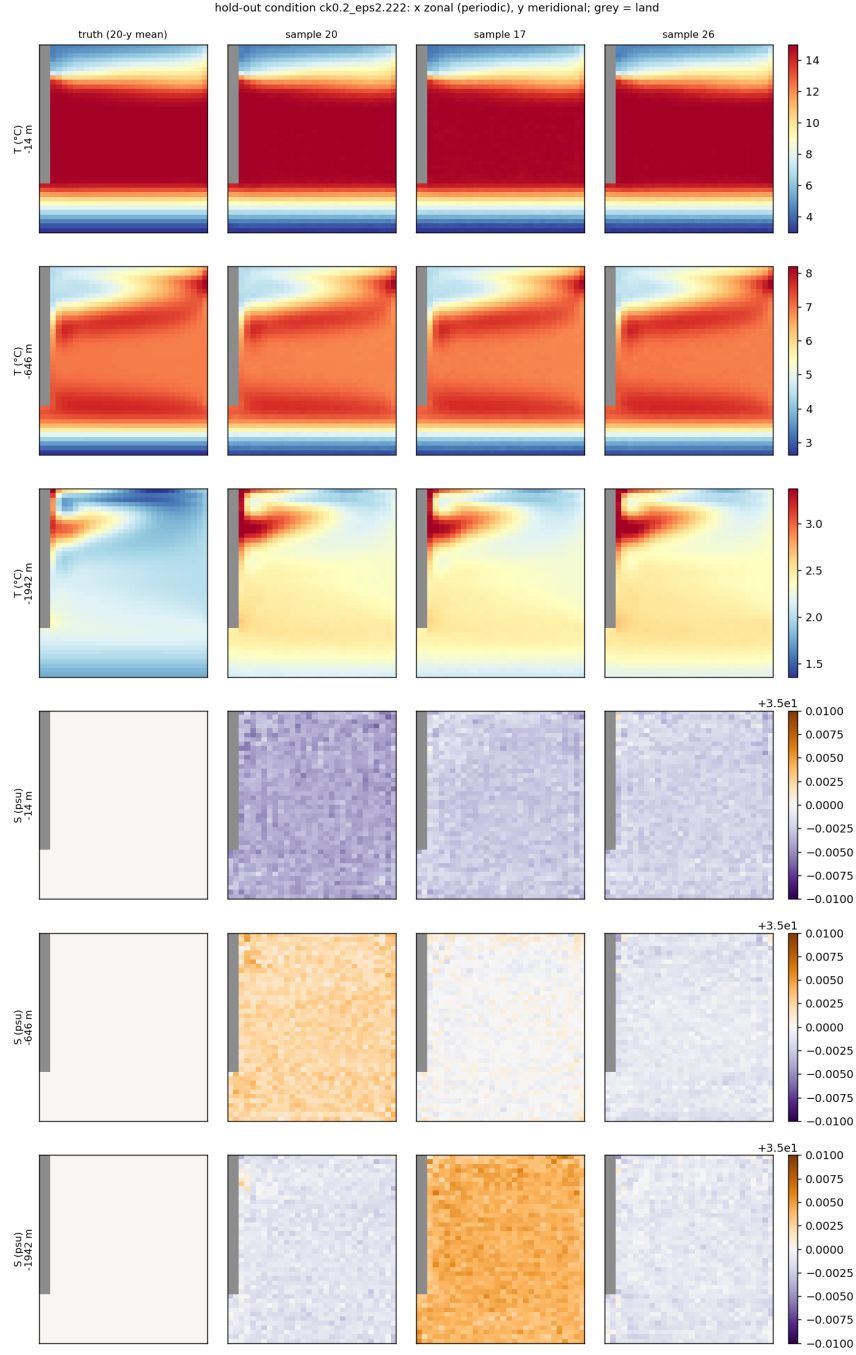


Figure 6: Band-of-three split, same condition ( $c_k = 0.2$ ,  $c_\epsilon = 2.222$ , now in the middle held-out row).

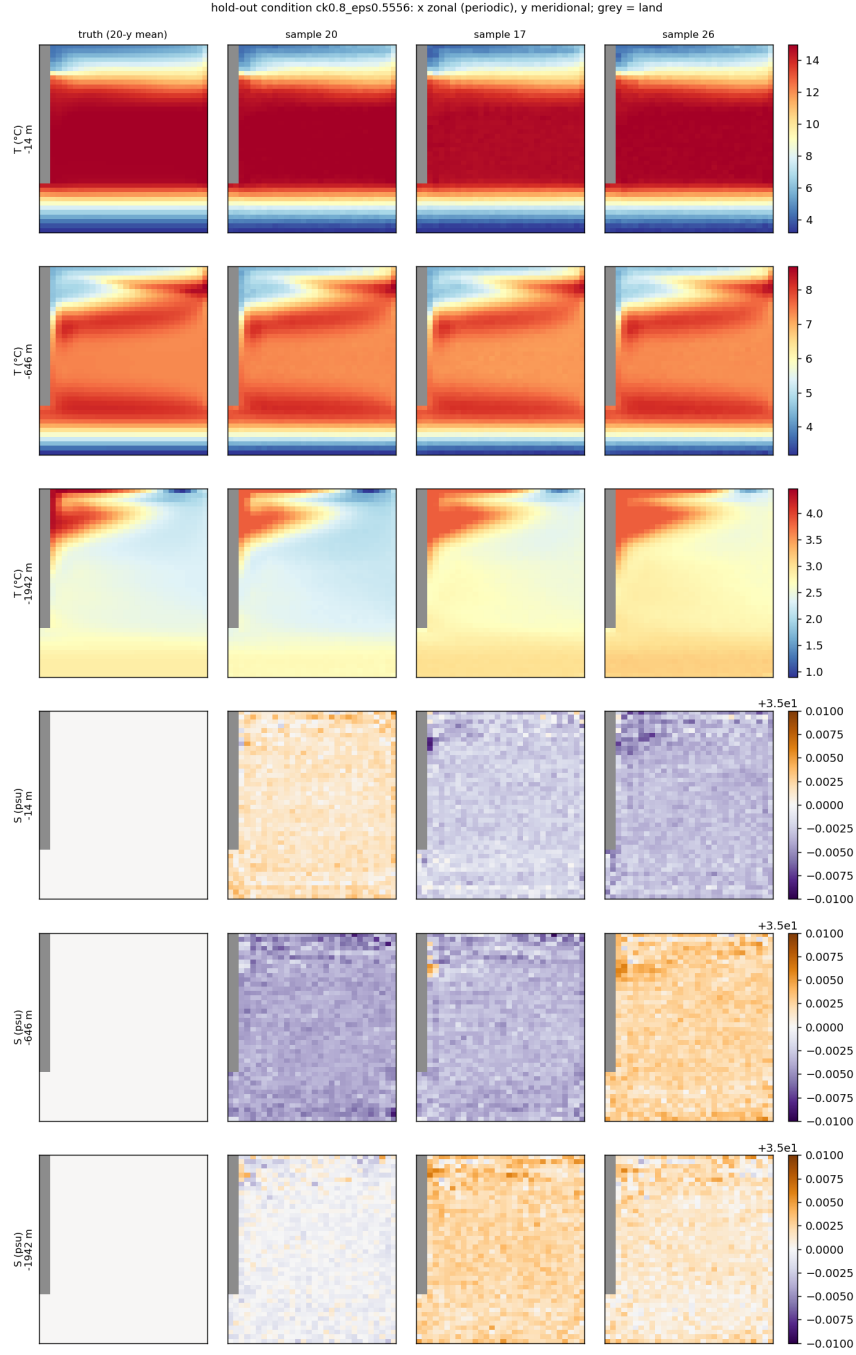


Figure 7: Top-row split,  $c_k = 0.8$ ,  $c_\epsilon = 0.5556$  (extrapolation one row above the training grid).